PRF7 - Dayend tables

PROFILE

Exported on 06/08/2021

Table of Contents

1 Table UTBLEVENT 4
2 Table UTBLEBCH 5
3 Table QUETBLEB 6
4 Table QUEUEB 7
5 Table QUEUEJ 8
6 Table QUEUEP 9
7 Table QUEUED 10
8 Table QUEUEC 11
9 Table SCATBL 12

The following tables are documented here:

- Table UTBLEVENT(see page 4)
- Table UTBLEBCH(see page 5)
- Table QUETBLEB(see page 6)
- Table QUEUEB(see page 7)
- Table QUEUEJ(see page 8)
- Table QUEUEP(see page 9)
- Table QUEUED(see page 10)
- Table QUEUEC(see page 11)
- Table SCATBL(see page 12)

1 Table UTBLEVENT

```
CREATE TABLE UTBLEVENT          -- Event Definition Table
  ( DESC T(40)                  -- (1) Description
  , EVENT T(12) NOT NULL        -- (1*) Event Name
  , PGM T(12)                   -- (2) Pre-processor Routine
  , PGM1 T(12)                  -- (3) Post-processor Routine

  , PRIMARY KEY EVENT
  )
```

The table has one additional computed column: STAT = $$STATUS^BCHMON(EVENT)

2 Table UTBLEBCH

```
CREATE TABLE UTBLEBCH          -- Batch/Event Definition Table
  ( BATCH N(4) NOT NULL        -- (2*) Batch Sequence Number
  , DES T(40)                  -- (1) Batch Description
  , EVENT T(12) NOT NULL       -- (1*) Event Name

  , PRIMARY KEY EVENT,BATCH
  )
```

Defines day-end "batches". Note the 4 digit numeric key.

If we assume a standard "supplier-part" model, the following foreign key definitions are missing:

- FOREIGN KEY (EVENT) REFERENCES UTBLEVENT
- FOREIGN KEY (BATCH) REFERENCES QUEUEB

3 Table QUETBLEB

```
CREATE TABLE QUETBLEB          -- Queued Event/Batch Submission Table
  ( BCHNUM N(4) NOT NULL       -- (2*) Batch Number
  , EVENT T(12) NOT NULL REFERENCES UTBLEVENT
                               -- (1*) Event Name
  , SUBM L(1) NOT NULL         -- (1) Queue Submission Indicator

  , PRIMARY KEY EVENT,BCHNUM
  )
```

(...)

4 Table QUEUEB

```
CREATE TABLE QUEUEB             -- Queuing Batch Level
  ( AOE N(1)                    -- (6) Action on Error
  , BCHNUM N(4) NOT NULL        -- (1*) Batch Sequence Number
  , CTIM C(11)                  -- (8) Actual Time Batch Completed
  , DESC T(40)                  -- (1) Batch Description
  , EVENT T(12)                 -- (18) Event Name
  , FRE F(12)                   -- (2) Frequency
  , LRD D(10)                   -- (3) Last Run Date
  , LSTAT N(1)                  -- (9) Status of Last Run
  , NRD D(10)                   -- (4) Next Run Date
  , REST L(1) NOT NULL          -- (14) Restrict Job Restore
  , STIM C(11)                  -- (7) Actual Time Started
  , STOP L(1) NOT NULL          -- (16) Stop Batch After Current Job
  , STRT N(3)                   -- (5) Offset
  , SUBM L(1) NOT NULL          -- (17) Queue Submission Indicator
  , VMSTAG T(16)                -- (15) Job Name for VMS Queuing

  , PRIMARY KEY BCHNUM
 )
```

Note that this table has an EVENT column which seems to conflict with the fact that table UTBLEBCH is keyed by BATCH and EVENT.

5 Table QUEUEJ

```
CREATE TABLE QUEUEJ             -- Queuing Job Level
  ( AOF N(1)                    -- (13) Action on Failure
  , BCHNUM N(4) NOT NULL        -- (1*) Batch Sequence Number
  , CDAT D(10)                  -- (16) Completion Date
  , CSTAT N(1)                  -- (6) Current Status
  , CTIM C(11)                  -- (5) Time Completed
  , DESC T(50)                  -- (1) Description
  , DIZ L(1) NOT NULL           -- (9) Delete Job From Batch When TTR=0
  , FUN T(20)                   -- (2) Function Name
  , JOBNUM N(3) NOT NULL        -- (2*) Job Sequence Number
  , LRD D(10)                   -- (3) Last Run Date
  , LSTAT N(1)                  -- (7) Status of Last Run
  , MAXR N(2)                   -- (11) Maximum Number of Resubmits
  , RETC N(2)                   -- (12) Retry Counter
  , RETI N(8)                   -- (10) Resubmit Interval
  , SDAT D(10)                  -- (15) Start Date
  , SKIP T(4)                   -- (14) Bypass Counter
  , STIM C(11)                  -- (4) Time Started
  , TTR N(2)                    -- (8) Times to Run

  , PRIMARY KEY BCHNUM,JOBNUM
  )
```

Defines the "jobs" with a batch. Jobs use a 3 digit number. The row contains a FUN column, which references SCATBL.FN. The additional condition is that SCATBL.QUEUE must be true.

Note that from the mapping of this table to a GTM database the following foreign key relationship is implicit:

- FOREIGN KEY BCHNUM REFERENCES QUEUEB

6 Table QUEUEP

```
CREATE TABLE QUEUEP            -- Queuing Parameter Level
  ( BCHNUM N(4) NOT NULL       -- (1*) Batch Number
  , JOBNUM N(3) NOT NULL       -- (2*) Job Number
  , PARAM T(40)                -- (1) Parameter
  , SEQ N(2) NOT NULL          -- (3*) Sequence Number

  , PRIMARY KEY BCHNUM,JOBNUM,SEQ
  )
```

Defines the batch job's parameters. SEQ starts at 1 and increments by 1. The PARAM value of the row with SEQ = N defines the value of the N-th input field in the UTLREAD screen associated with the function.

7 Table QUEUED

```
CREATE TABLE QUEUED             -- Queuing Dependency Level
  ( BATCH N(7) NOT NULL         -- (3*) Index Batch Number
  , BCHNUM N(4) NOT NULL        -- (1*) Batch Number
  , BCHPRMT N(7)                -- (4) Batch Number Prompt
  , CDATE D(10)                 -- (2) Completion Date
  , CTIME C(11)                 -- (3) Completion Time
  , JOB N(3) NOT NULL           -- (4*) Index Job Number
  , JOBNUM N(3) NOT NULL        -- (2*) Job Number
  , JOBPRMT N(3)                -- (5) Job Number Prompt
  , LIMIT N(8)                  -- (1) Dependency Time Limit

  , PRIMARY KEY BCHNUM,JOBNUM,BATCH,JOB
  )
```

The table documentation does not specify in which direction the dependency is defined. According to others a row in this table means that (BCHNUM, JOBNUM) must wait until (BATCH, JOB) is finished.

Notes:

- Column BATCH is defined as N(7) where other batch numbers are N(4).
- Based on the table-to-global mapping the following constraint is implicit: FOREIGN KEY (BCHNUM, JOBNUM) REFERENCES QUEUEJ
- The following foreign key definition is missing from the schema as well: FOREIGN KEY (BATCH, JOB) REFERENCES QUEUEJ

8 Table QUEUEC

```
CREATE TABLE QUEUEC             -- Queuing Dependency Level
  ( BATCH N(7) NOT NULL         -- (3*) Index Batch Number
  , BCHNUM N(4) NOT NULL        -- (1*) Batch Number
  , JOB N(4) NOT NULL           -- (4*) Index Job Number
  , JOBNUM N(4) NOT NULL        -- (2*) Job Number

  , PRIMARY KEY BCHNUM,JOBNUM,BATCH,JOB
  )
```

Although implemented as a separate table, this structure is nothing but an index on QUEUED with the column names "reversed".

Notes:

- Column BATCH is defined as N(7) where other batch numbers are N(4).
- Columns JOB and JOBNUM are both defined as N(4) whereas in all other tables they are defined as N(3)
- Based on the table-to-global mapping the following constraint is implicit: FOREIGN KEY (BCHNUM, JOBNUM) REFERENCES QUEUEJ
- The following foreign key definition is missing from the schema as well: FOREIGN KEY (BATCH, JOB, BCHNUM, JOBNUM) REFERENCES QUEUED

9 Table SCATBL

```
CREATE TABLE SCATBL             -- System Function Table
  ( %LIBS T(20)                 -- (10) Library Name
  , %SN T(12)                   -- (11) System Name
  , BPSFLG L(1) NOT NULL        -- (17) Block Mode Support Flag
  , BREAK L(1) NOT NULL         -- (15) Restrict Break
  , DDP L(1) NOT NULL           -- (5) Valid DDP Function
  , DESC T(50)                  -- (1) Function Description
  , FN T(12) NOT NULL           -- (1*) Function Name
  , NOHOST L(1) NOT NULL        -- (16) Invalid Host Function
  , NOREPOST L(1) NOT NULL      -- (19) Exclude from Repost
  , PGM T(200)                  -- (4) Program Name
  , POP T(255)                  -- (3) Post Processing
  , PRP T(255)                  -- (2) Pre Processing
  , QUEUE L(1) NOT NULL         -- (18) Valid to Queue
  , RESTORE L(1) NOT NULL       -- (14) Restrict Restore
  , SALON L(1) NOT NULL         -- (6) Valid Stand Alone
  , TIMBEG C(10)                -- (7) Valid Time Beginning
  , TIMEND C(10)                -- (8) Valid Ending Time

  , PRIMARY KEY FN
  )
```

Strictly speaking, this this table is not a "day end" table. However, in the end the queuing system is nothing but a system to run functions from this table in sequence.

Whether the function is valid for execution by the queuing system is determined by SCATBL.QUEUE. When a valid function is executed as a job in a queue by JOBNUM1^QUEPGM(), this method will xecute SCATBL.PRP, do @SCATBL.PGM and xecute SCATBL.POP.